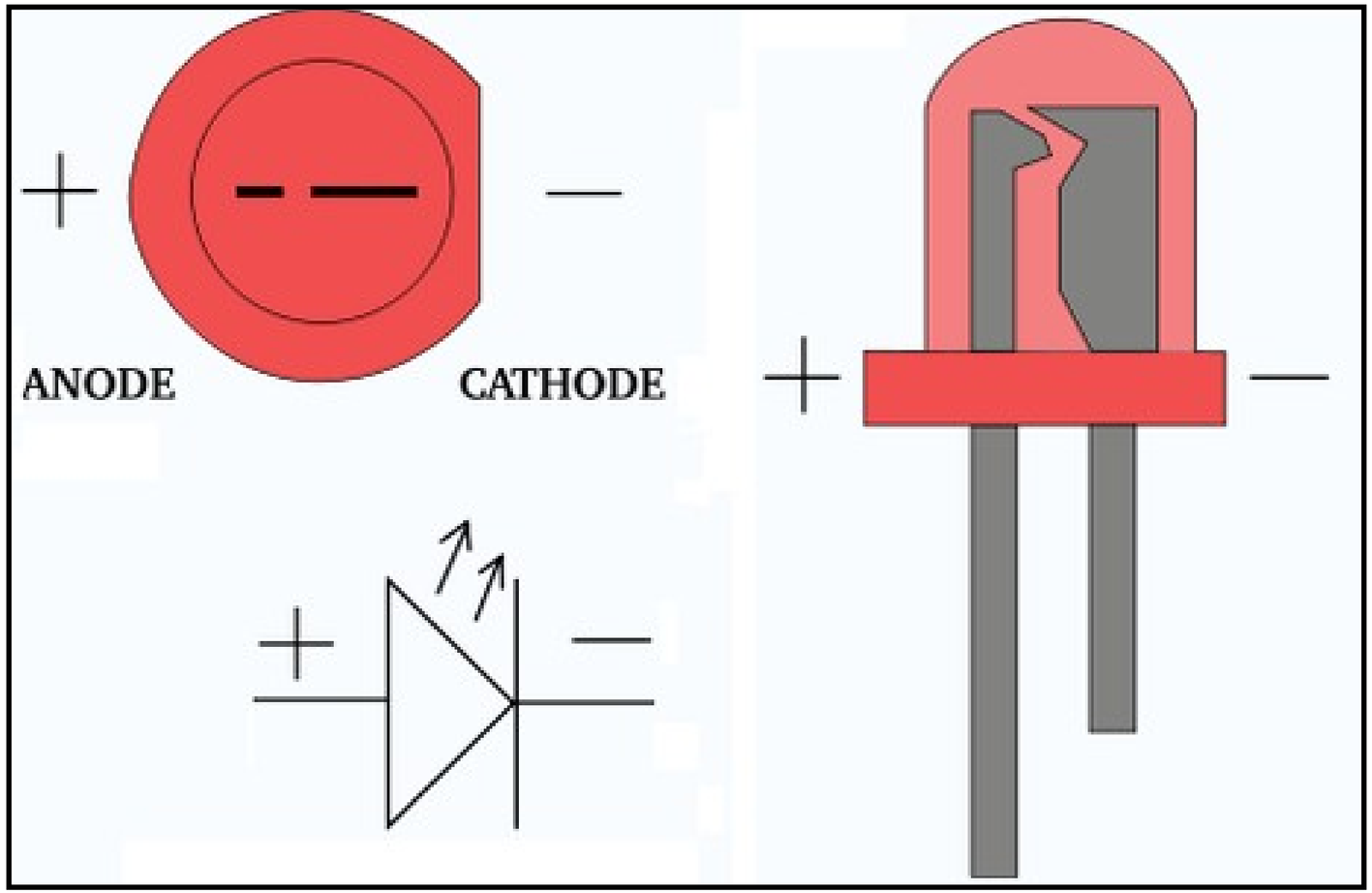
Health Monitoring System

Interfacing with ESp32

- **LED**
- **Temperature sensor**
- **Pulse oximeter**

Basics of LED



1.Blinking of LED

i. On-board LED blinking

code:-

```
/*ON Board LED GPIO 2



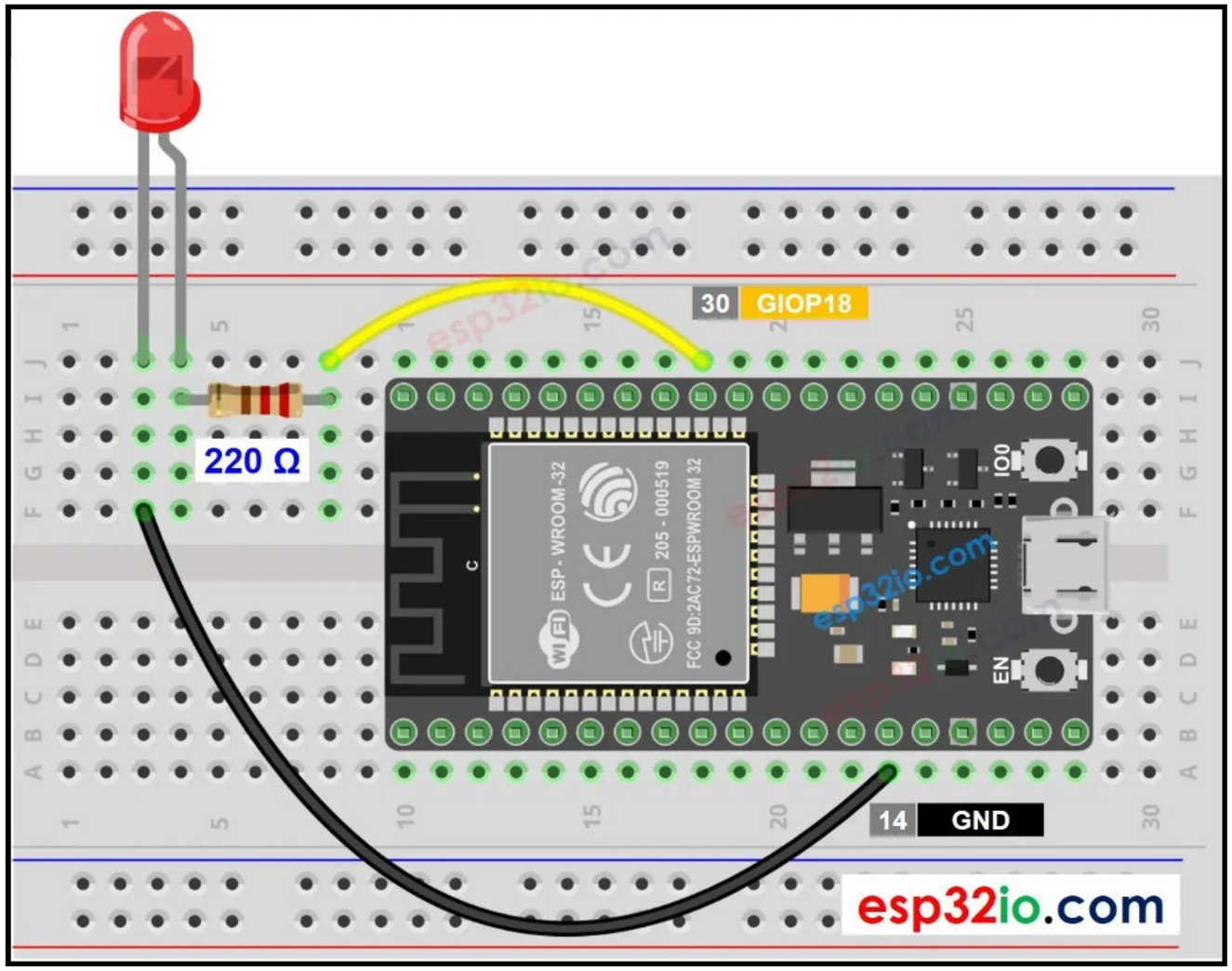
#define LED 2

void setup() {
  // Set pin mode
  pinMode(LED,OUTPUT);
}

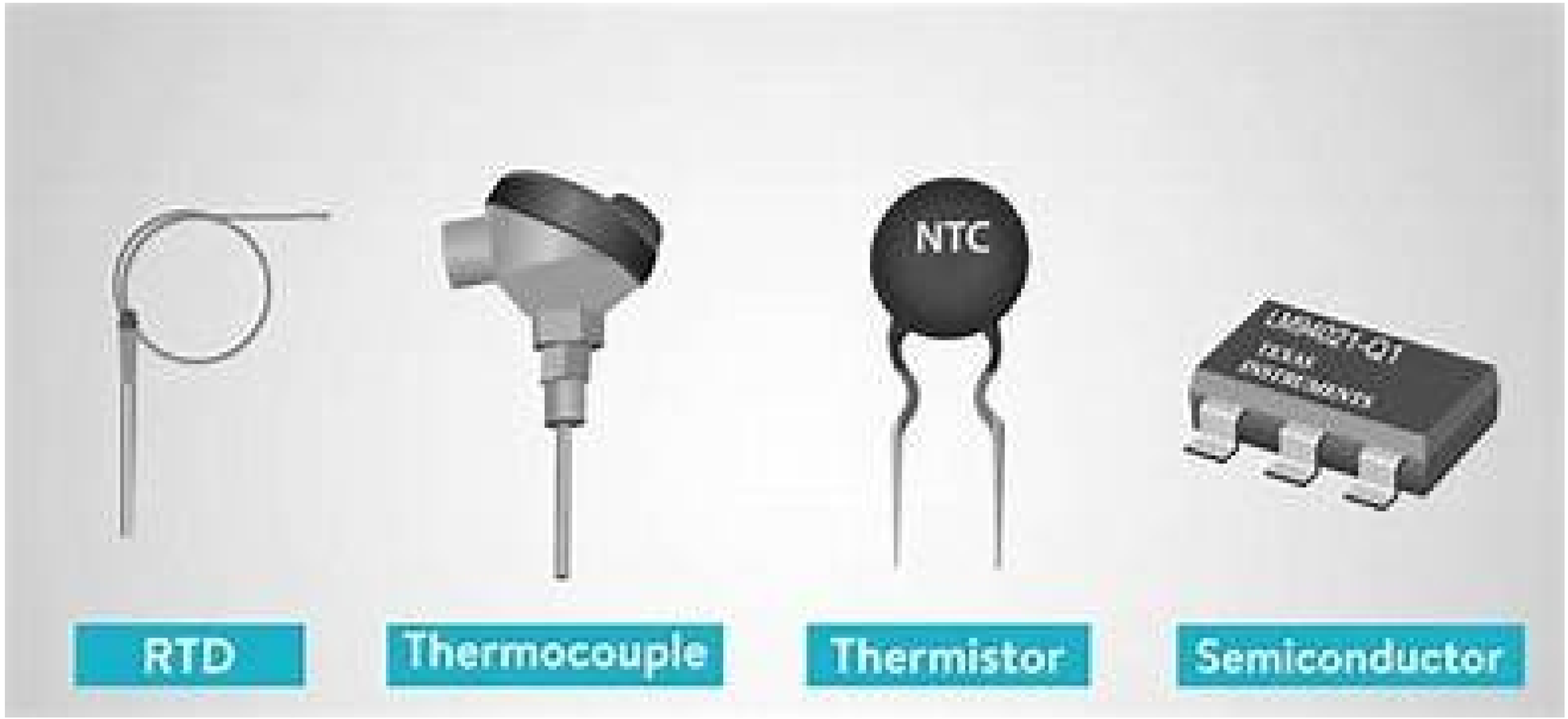
void loop() {
  delay(500);
  digitalWrite(LED,HIGH);
  delay(500);
  digitalWrite(LED,LOW);
}
```

ii. Off-board LED blinking

CIRCUIT DIAGRAM

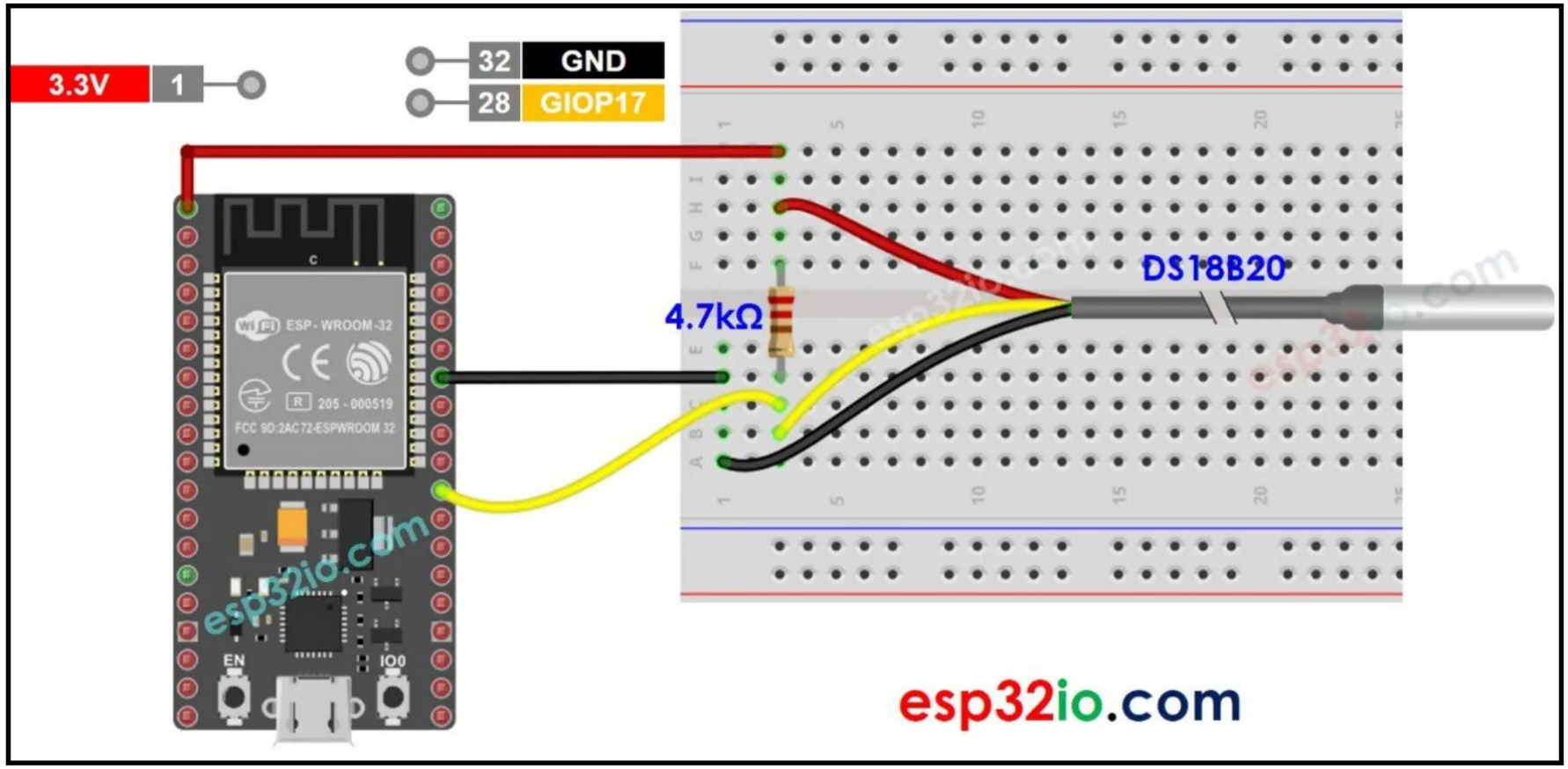


Types of Thermal Sensor



2.Interfacing Temperature sensor

CIRCUIT DIAGRAM:-



CODE:-

```
#include <OneWire.h>
#include <DallasTemperature.h>

#define SENSOR_PIN  26 // ESP32 pin GPIO17 connected to DS18B20 sensor's DATA pin

OneWire oneWire(SENSOR_PIN);
DallasTemperature DS18B20(&oneWire);

float tempC; // temperature in Celsius
float tempF; // temperature in Fahrenheit

void setup() {
  Serial.begin(9600); // initialize serial
  DS18B20.begin();    // initialize the DS18B20 sensor
}

void loop() {
  DS18B20.requestTemperatures();       // send the command to get temperatures
  tempC = DS18B20.getTempCByIndex(0);  // read temperature in °C
  tempF = tempC * 9 / 5 + 32; // convert °C to °F

  Serial.print("Temperature: ");
  Serial.print(tempC);    // print the temperature in °C
  Serial.print("°C");
  Serial.print("  ~  ");  // separator between °C and °F
  Serial.print(tempF);    // print the temperature in °F
  Serial.println("°F");

  delay(500);
}
```

IOT based using BlynkIOT

